Fiscal Year:	FY 2018	**Task Last Updated:**	FY 04/02/2018
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	12/31/2021
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2021 per U. Israelsson/JPL (Ed., 1/8/19) NOTE: End date changed to 12/31/2018 per PI (Ed., 3/10/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1459238		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ACES (Atomic Clock Ensemble in Space) is a European Space Agency (ESA) project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for at least 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We will assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The current working uncertainty budget projects 2.2×10^(-16) accuracy. The proposed work will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of the others, with a goal of reaching 10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify workable procedures to demonstrate its accuracy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on microwave cavities produced an improved design that will be used for Cs and Rb fountain clocks. These are being incorporated into new clocks being built at the British, South Korean, Polish, Canadian, and Indian National Laboratories. The new cavity design allows significantly reduced clock errors due to Doppler shifts. Our work has also led to a deeper understanding of the microwave lensing frequency shift that is being used as a correction of International Atomic Time and, more broadly, our work contributes to scientific progress towards anticipated future optical frequency atomic clocks.
Task Progress:	The primary focus of our recent work on the accuracy evaluation of PHARAO was optimizing the on-orbit sequence to enable the ACES time transfer sciences goals and the PHARAO accuracy goals. The ACES time transfer goals--to test general relativity, to probe the time variation of fundamental constants, and to compare high-performance ground clocks worldwide--benefit from PHARAO operating for approximately 10-day periods with its highest stability that is consistent with its accuracy goals. To optimize the stability and accuracy, the impact of the collision shift has to be managed, as well as the evaluation of the cavity phase shifts. Our approach gives a new solution for the accuracy evaluation of these errors, and is expected to allow PHARAO to achieve its accuracy and stability goals. To address the systematic errors of the evaluation of the PHARAO collision shift, we developed a Monte Carlo simulation, built on our previous simulations for our cesium precision scattering measurements. We used the collision shifts from the latest and significantly improved cesium-cesium interaction potentials from our theory collaborations. Our results nominally agreed with previous simulations from SYRTE and showed the expected behavior at high launch velocities, where the previous simulations gave a shift that decreases. Our Monte Carlo incorporated the inhomogeneities of the state selection, state detection, and the tipping angles of the Ramsey pulses. The inhomogeneous microwave field amplitude in the state selection cavity changes the density distribution when ¼ versus all of the atoms are selected. This introduces a systematic error into the evaluation of the collision shift, since the frequency shift is no longer linearly proportional to the number of detected atoms. Our knowledge of the amplitude gradient of the microwave field in the selection cavity is highly uncertain, with more measurements required. The current best fit gives a systematic error of 1×10^-17, which is sufficiently small, provided the selection amplitude gradient is not much larger than the current estimate. If the amplitude gradient in the state selection cavity is larger, it may require using detuned state selection on orbit, as opposed to simply changing the microwave amplitude. A significant advance was developing a new operating procedure for PHARAO that allows both a 10^-16 accuracy and operating with a high stability for dedicated time transfer periods. The previous scheme was to continuously perform density and launch velocity extrapolations, but these required 16 months of averaging to reach 10^-16 and significantly degraded the PHARAO's stability. Instead, by operating PHARAO with slightly different parameters for the density and launch velocity extrapolation, the Doppler shift correction at a particular launch velocity can be measured to 5x10^-17 in a significantly shorter time, of order 4 months. This procedure relies on the cavity phase error being small, as expected from ground measurements, and this error will be confirmed on orbit.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Articles in Peer-reviewed Journals	Bennett A, Gibble K, Kokkelmans K, Hutson JM. "Atomic clock measurements of quantum scattering phase shifts spanning Feshbach resonances at ultralow fields." Phys Rev Lett. 2017 Sep 15;119(11):113401. https://doi.org/10.1103/PhysRevLett.119.113401 ; PubMed PMID: 28949205 , Sep-2017